Home › My Courses › Financial Markets › M7: Integrating Ethics with Financial Challenges › Lesson Notes

Lesson Notes

MODULE 7 | LESSON 2

KEY CONCEPTS IN FINANCIAL MARKETS

Reading Time	90 minutes
Prior Knowledge	Comprehensive Modules 1 thru 6

In the previous lesson, we compared how securities get priced in normal markets and then the challenges that throw these methods off course. In this and the next lesson, we review the first 10 sets of key financial concepts: collateral, statistical, magnifying, frictional, and failure. These concepts affect the data we use, the models we build, and the decisions we make when we are solving problems that involve pricing, hedging, and allocating funds. We will review these concepts in pairs and see some examples that combine several concepts at once.

1. Collateral-Related Concepts: Credit Risk and Financing

1.1 Long Trades vs. Short Trades

Remember that in financial markets, investors can enter a trade in two distinct ways. They can purchase a security they do not own—-this is known as _going long—-_or they can borrow a security they do not own from someone who does own it and sell it—this is known as *going short*.

- When the investor is long, they anticipate the price of the security increases so that they can sell the security at that higher price. When they do so, they close the position and find themselves with a profit as the net result of the two transactions. The plan is to "*buy low, then sell high*."
- When the investor is short, they anticipate that the price of the security will decrease. The investor hopes that they can buy the security at a lower price later. When they buy it back, known as _covering the short, _they receive the security, return it to the party that loaned it to them, and close their position. The plan is "*sell high, then buy it back low*."

What does an investor need in the situation where they go long? In a word, funds. If this is an individual, they need to have cash in their account. If this is an asset manager, they need to have the funds of their clients. Of course, what if they do not have enough funds? They can finance by borrowing money, paying for a portion of the securities with borrowed funds and the rest with actual funds. Long positions have the flexibility of using financing or not, depending on whether the investor wants to be more conservative (using all cash) or invest more aggressively (using financing/leveraging).

What does an investor need in the situation where they are short? They need to borrow the security. Who loans it to them? Someone who owns that security! Well, why do you think someone would loan a security to another investor with whom they neither met nor have financial relationships? The answer is income: the owner of the security will be paid a financing fee. Thus, financing is not free. Anything borrowed has an associated cost. This cost is assumed to be different things in different models. In Black-Scholes, this cost is assumed to be zero. That is not a great assumption. In a paper by the late quant Marco Avellaneda called "Option Pricing for Stocks that are Hard-to-Borrow," the cost is assumed to be a Brownian motion that can vary with the supply and demand forces for that security. This model for financing costs is more realistic and can lead to better pricing in the options market.

2. Credit Risk and Financing

Credit risk and financing are the first set of challenges we address in financial engineering. Credit risk has been defined several times in different places in our course. Think of credit risk in simple terms: a lender gets less money back than what was originally lent or expected back. Recall from the first few lessons you saw credit risk from the bank's side (in a mortgage) and from the depositor's side (with the lender). In 2022, the failure of FTX to secure funds of investors illustrates the credit risk that exists when funds from brokerage accounts are lent out and not returned. The party that is owed money likes to keep track of their counterparty's credit risk. When collateral is in place, the problem extends to financing and a calculation of market risk (of that collateral). We discussed how collateral, regulation, and insurances (like the FDIC) help to mitigate credit risk.

In subsequent modules, you read that lending rates are a function of several key factors, three of which include the term of the loan, the creditworthiness of the borrower, and the type of securitization, if any, tied to the loan. During markets where the yield curve is upward sloping, the rates for short-term loans tend to be lower than the rates for long-term loans. The rates for securitized loans are lower than the rates for un-securitized loans. For example, home mortgage interest rates are lower than consumer credit card rates. If a homeowner defaults, the bank can take the home as a substitution. If a credit card borrower defaults, the bank may have nothing tangible left—the funds could have been spent on massages, products, and services that were consumed or have lost significant value. Finally, the rates for better creditworthy individuals are more favorable than those borrowers that are considered high risk.

2.1 Examples of Credit Risk

The ability to model credit risk allows the financial engineer to solve a vast set of problems. How do you price a corporate bond? Given a borrower's credit history, what is a suitable interest rate for a securitized loan? What would BBB bonds expect to recover if they were to default? These questions require us to understand the key variables of credit risk.

Let's take the corporate bond example. We'd start by pricing a bond that cannot default and then, somehow, include the risk of default. In credit risk, we have various ways that you can measure the risk of default. This is known as the probability of default. We might get these by looking at stocks, credit default swaps, credit ratings, or other factors we build. Therefore, we need financial data and models. For example, sometimes, we'll use the capital structure of a company to model default probability. We'll want to know the company's assets and debts. Then, we'll model its equities like an option. You saw that in Module 4. We'll expand upon it in our discussion of options.

Credit risk is influenced by, and influences, asset classes during times of market stress. Higher default probabilities or lower recovery rates are two examples of credit risks that can increase. Higher default probabilities will increase funding costs, raise debt-to-equity ratios (discussed in Module 5), force higher collaterals to be set aside as cushions for unexpected losses, and make it more difficult for a firm to be profitable. As markets get more stressed, this in turn nudges higher defaults, lower recovery rates, and greater expected losses. For example, consider a borrower with a variable-rate mortgage. Suppose interest rates increase. In turn, this will likely increase mortgage rates. With a variable-rate mortgage, the costs of the mortgage have increased. Suppose the borrower is no longer able to afford these higher payments. At the same time, higher interest rates and mortgage rates also caused home prices to drop, as the higher rates decrease the demand for housing. The value of the collateral—*the house—is now worth less!* So there are two credit risks that worsened: the default probability increases and the loss given default increased, since the collateral is now worth less.

In credit risk, there's several variables involved in the expected loss: default probability, loss given Default, and the exposure at default. The first two of these are correlated. We're not escaping correlation. We're just doing one set of factors each week. But you can see how that can lead to what's known as different ways that you would model an expected loss.

2.2 Building the Credit Risk Toolkit

Credit risk has real-world practicality. Personally, you may decide you want to lend a friend money. Somehow, you may run a quick model in your head: "Have I already lent them money? Have they paid me back? Are they likely going to pay me back? Are the funds used for something serious?" Credit risk is something that most people can understand intuitively.

So what do we do with credit risk?

- In Econometrics, we'll look at regression models, such as logistic regressions, where we can model a default as a Poisson process. We will get exposed to different distributions that are suitable to model default probability and loss.
- In Derivative Pricing, we'll understand default as a probability like a European call option ending up out-of-the-money. We'll also learn new products like the credit default swap that isolates credit risk, making it different than the econometric approach.
- In Stochastic Models, we'll examine examples of calibrating credit default swaps to imply a default probability. We'll expand our mathematical toolkit to include Markov processes to allow us to model the migration of credit transition matrices.
- In Machine Learning, we'll use supervised learning to improve upon models that predict default.
- In Deep Learning, we'll provide more detailed ways we can back into credit risk variables from observed credit derivative products like first-to-default baskets.

- In Portfolio Management, we will measure and manage credit risk in examples where we use copulas to model joint defaults.
- In Risk Management, we'll use a network approach in Bayesian networks to model credit risks.

Clearly, credit risk is a recurring theme throughout the whole program. There is no escaping credit risk, particularly since the global financial crisis when so many experts got credit risk wrong. Since the GFC, even something as simple as pricing an interest rate swap now requires credit risk calculations, known as XVA adjustments, that credit or debit value based on the credit risk of you and your counterparty's credit risk.

3. Financing

3.1 Examples of Financing

Financing reflects the cost of borrowing something. Clearly, it is related to credit risk. Suppose you short a security. Step one is that you borrow it from them. The owner charges a fee—that's where financing enters. Owners of securities who lend them out do so to make extra return. Let's say that you owned Meta stock (formerly known as Facebook). You agree that you are willing to lend your shares to earn some extra income (income in addition to any dividends it would pay). Other people may be bearish about the stock and believe it's overpriced, so they may want to sell the stock. But they currently don't own it. They can borrow shares from you. You would then charge whatever the market rate is, let's say it's twenty basis points (meaning 0.20% interest on an annualized basis). Once they have the stock, they sell it short in the market. They're hoping the stock price drops, so they buy it back (cover) at a lower price. If that becomes true, then they've made money by selling high and buying low. However, the price could increase, and they may have to cover at a higher price. Either way, the borrower had to pay a financing cost to the original owner. Securities are not library books. You could go to your local library and presumably borrow a book for free and return it two weeks later. That's not how securities work. If you went to a brokerage and said, "I want to borrow shares of Meta," whoever owns those shares would agree when they open their account to lend those securities, and that would earn them extra income. Suppose Meta paid a dividend. Then, the borrower who borrowed the stock also has to pay the dividend in addition to the financing cost. The true cost of borrowing is a combination of the income it produces (that is, the dividend) and the financing cost, volatile as it may be.

3.2 Building the Financing Toolkit

- In Econometrics, we'll see how the cost of carry can be modeled as a time series that depends on certain exogenous variables.
- In Derivative Pricing, we'll look at a very simple financing model (hint: no cost) and how you hedge a put by shorting the underlying with the Black-Scholes model. However, if we just used these assumptions, we would never reproduce the volatility smile.
- In Stochastic Models, we'll look at a model of how you can include financing costs in the hedge. For example, we'll model the cost of financing as an additional Brownian motion and show how this model reproduces the real-world option prices better.

- In Machine Learning, we'll examine financing associated with performing a long-short strategy, where you buy one security and sell short another, with the objective that the long security outperforms the short security.
- In Deep Learning, we'll extend some previous examples to use more market data.
- In Portfolio Management, we'll use financing costs as part of constraints in minimizing costs to rebalance a portfolio.
- In Risk Management, we'll discuss liquidity crises that affect financing costs.

4. Understanding the Motivation of the Curriculum

Before we look at the next set, fast forward two years from now. As you're worked through the program, a few of these factors have become your favorite problems. For example, if you enjoy solving problems in credit risk, you could certainly find work in that area. Within the field of credit risk, there are specializations: counterparty credit risk modeling; credit rating agency factor modeling; credit derivative trading; corporate bond portfolio management; structured credit sales; credit risk management; and so on. When you're getting assessed later in an interview, think about the tools you learned. Rather than thinking, “what did I study in Econometrics?” you can ask yourself “what skills did I build in Econometrics that I applied to solve credit risk problems?”

So you may wonder, why not have one course dedicated to each of these areas, like credit risk and volatility modeling, instead of courses like Econometrics and Machine Learning? This relates to the difference between theory and practice. Econometrics has foundations in linear algebra, probability, statistics, and calculus. These are used to build models. Once you have these statistical, mathematical, and machine learning models, you can apply them in many ways to solve problems in credit risk, financing, leverage, regulation, etc. The educational process in engineering could not possibly teach you every application. That's its strength and beauty. Instead, we apply the models to solve problems in these areas of estimating credit losses, measuring volatilities, and more. Some courses will emphasize a particular factor. For example, Derivative Pricing will illustrate the power of both leverage and non-linearity. Others can be broadly applied to any areas. Machine Learning can tackle problems across the spectrum.

You are encouraged to think of each metric, methodology, or model as a tool, as if you were a carpenter. Imagine you are training to be a carpenter. You're going to develop skills in modeling, measuring, cutting, adhering, smoothing, etc. To perform these skills, you would learn to use a variety of tools. Likewise, think of these areas as opportunities for skills development: skills for handling credit risk, addressing financing, etc. As you work through the curriculum, think about how the material illustrates how a tool is used, teaches you to use that skill, and builds a foundation to understand its justification. Don't miss the forest for the trees! Instead of overemphasizing technicalities, focus on the big picture. Ask yourself, "how did this course improve my mathematical, statistical, or computational background to better understand, model, predict, or interpret credit risk? Volatility? etc."

5. Statistical Factor 1: Volatility

A big part of the Financial Markets course addressed volatility and correlation. Volatility means that we have uncertainty around an expected value. We expect a certain return, a certain payoff, a set of cash flows, etc., but we often get different results. Volatility describes that expected difference.

Considering more than one security at at time, we have a covariance—how two or more securities vary together. This depends not only on their individual variances but also on their correlation. Correlation is the degree to which two securities move together. Of all these financial challenges, most of the attention in your studies will focus on ways to measure and manage volatility and correlation. Let's define volatility and then discuss a dozen of its properties.

- Volatility can mean risk or uncertainty.
- Volatility is indifferent to direction.
- Volatility is nonlinear.
- There are different types of volatilities.
- There are different ways to measure volatility.
- Securitized portfolios have securities with similar variances.
- Volatilities change over time.
- Volatilities do not necessarily scale with time.
- Volatilities can be implied from derivatives, especially volatility derivatives.
- Volatilities affect other variables.
- Variances are the diagonal of the covariance matrix.

5.1 Volatility can mean risk or uncertainty.

Volatility is often used synonymously with risk, a term people use often: risk management, risk aversion, risk premium. Risk typically describes situations where the outcomes—their distributions and probabilities—are known. Let's say we're flipping a coin many times. If all the events are independent and we run lots of them, the long-term results should converge to a normal distribution. Then, we could characterize the risk through metrics like Z scores, which rely on standard deviations. The calculations and methods you know and use from statistics find a home when discussing risk. Although we don't know what a specific outcome is, we have general comfort in knowing what the distribution is. When you studied statistics, you learned of these different distributions. Some were Gaussian. Others were skewed or kurtotic. That is, many have heavy tails compared to the normal distribution. The heavier the tails, the greater the chance of an extreme outcome (assuming we don't have disproportionately more chances of zero outcomes). In simpler terms, volatility is more uncertain in heavy-tailed distributions. For example, in 2022 Q4, the New York Stock Exchange set a record: the market moved down and then up so much that the total distance covered was more than ever before. Although there have been larger moves in one direction on a given day, the record was the comeback within a single day. This reflects very high intra-day volatility. It means that the distribution that we have may not be Gaussian.

One thing that makes volatility difficult is distinguishing risk (known distributions) from uncertainty (unknown distributions and/or outcomes). How do we know that we have the right distribution? If we had some understanding of the shape and the value of the parameters, then we're using classical or

Bayesian statistics to draw inferences. For example, parametric Value-at-Risk (VaR) requires a distributional assumption; then, we pick a percentile and report that losses do not exceed a specific value, say, 99% of the time. When we are unsure of the distribution, or even the outcomes and probabilities, we switch the terminology from risk to uncertainty. Perhaps the return of Bitcoin falls into this category. Uncertainty means there's less consensus (if any) on the underlying distribution(s). We may have unstable parameters, changing shapes, or mixtures of distributions. For example, you may have a distribution known as a mixture of normals—a combination of 2 (or more) normals, each with its own mean and variance, plus a proportion of how they mix. Perhaps the second normal has the same mean, but a much larger standard deviation. You put these together, and you have a mixture of normal distributions that introduces some of the heavy-tailed nature that you'll see. The parameters of the mixture (for the bivariate normal there are 5) may be time-varying.

Risk and uncertainty describe what you'll see in this whole master's program as volatility. In the Financial Markets course, Modules 2 and 3 presented to you two different asset classes. You saw both cryptocurrencies and stocks. Those were chosen because some asset classes are a lot more volatile than others. Even within stocks, you have some stocks that are much riskier than others because they are companies that are younger or perhaps just belong to industries that inherently have more uncertainty.

5.2 Volatility is indifferent to direction and semi-deviation.

Indifference to direction can be interpreted by saying that if the order of the elements is changed, their standard deviation does not change. In other words, we could organize the data in ascending or descending order and the value of the volatility would not change. This uncertainty is indifferent to direction. When risk increases, there are greater chances of moves in either direction. Volatility relates to the size of a move rather than its sign. Consider a random variable that is bounded; for example, default probabilities lie between 0 and 1 inclusive. Suppose the current value is 1%. Increasing volatility will mean it can move in either direction, but to be sure the boundary at 0 means there is much less room to move on the downside than on the upside.

However, we could also interpret "*indifference to direction*" as a lack of distinction between "good" and "bad" volatility. For instance, assume that an investor buys shares of XYZ at a price of \50/share. Then, suppose that the price keeps going up almost daily and that 1 year later the shares of XYZ are valued at $120/share. The stock has exhibited a good amount of volatility during the last 12 months, but none of the investors who bought the shares of XYZ would be upset. On the other hand, if the stock kept going down and 1 year later it stood at, say $15 per share, investors in XYZ would be very upset. Volatility does not distinguish the good direction from the bad one and if were were to assess risk from the volatility alone, we might conclude that the investment was riskier when it went from $50/share to $120/share then it was when it went from \50/share to $15/share. Clearly, that does not seem to cut it.

For this reason, portfolio managers, besides regular volatility, may compute what is known as **semi-deviation**. The downside semi-deviation measures the volatility or dispersion of negative returns below a specified threshold. Let Brepresent the desired target return; the the semple target semi-deviation formula is given by

$$s_B = \sqrt{\sum_{r_i \leq B} \frac{(r_i - B)^2}{N-1}},$$

where N is still the number of observations in our sample, but the sum considers only those returns, r_i 's that are below the required threshold B.

Example. *Fund MNO over the last 10 years have achieved the following returns*:

$$4.5\%, 6.0\%, 2.2\%, -2.0\%, 0.0\%, 5.2\%, 4.1\%, 2.5\%, 5.5\%, 4.3\%.$$

- *Compute the standard deviation of returns for fund MNO over the given 10 years*;
- *Compute the semi-deviation of returns over the same period for fund MNO if the target return is* 2%.

Answer. The standard deviation can be computed using the usual formula for the sample standard deviation and it is equal to 2.57%.
To compute the required semi-deviation, we need to use the formula given above with $B = 2\%$, and $N = 10$. Then, since only the returns less than or equal to 2%, matter, we find

$$s_B = \sqrt{\frac{1}{9}[(-2.0\% - 2.0\%)^2 + (0\% - 2\%)^2]} = 1.49\%.$$

The notion of semi-deviation just introduced focuses on the volatility of returns below a specified threshold or target return and allows us to better estimate the likelihood of experiencing negative outcomes.

5.3 Volatility is non-linear.

Volatility is non-linear. If we were to combine the volatilities of 2 securities, it is not simply their weighted average. Correlation affects the total. We'll see this explained in a few points.

5.4 There are different types of volatility.

Sometimes, the word volatility will have another term in front of it. You will see historical volatility, realized volatility, implied volatility, local volatility, stochastic volatility. Some of the distinction is due to the securities used to calculate the volatility. Other distinctions are based on the type of model used.

5.5 There are different ways to measure volatility.

Volatility can be measured with different metrics: variance, semi-variance, inter-quartile range. It may use different data that results in different metrics. For example, using prices we can compute the price range, high price minus low price. Using returns, we can compute variances.

5.6 Securitized portfolios have securities with similar variances.

Recall in discussions of products like MBSs, the securities should have similar risk characteristics. This would facilitate modeling their prepayments. Imagine if the mortgages had very different maturities, interest rates, types (fixed or variable). It would be incredibly difficult to model the overall MBS prepayment. Securitization relies on having securities with similar risk profiles.

5.7 Volatilities change over time.

Sometimes, we'll assume volatilities are constant. (See the Black-Scholes model.) As we develop stronger modeling skills, we'll assume volatilities change. GARCH, by definition, models a changing variance: the H represents heteroskedasticity, from the Greek words for "different variances." In modeling option prices, stochastic differential equations will provide more flexibility to include variance as an additional source of variation aside from the underlying. One of the key choices we'll make in any model is which quantities are parameters versus which ones are variables.

5.8 Volatilities do not necessarily scale with time.

Volatility depends on the timescale on which it was measured. One thing that you really need to understand early and appreciate as you start collecting data and think about models is at what frequency you're measuring data. Most of you will start by looking at daily closes: take the end-of-day price. That's well defined because markets have an official closing price. Some markets trade 24/7 (e.g., Bitcoin), so there is no close. Perhaps you trade high-frequency and need to examine volatility within minutes or seconds. The key point here is that volatility over, say, 1 minute, is a neat fraction of the volatility for a longer period. Sometimes, you may assume volatility scales like a Gaussian, but this is often not advised.

5.9 Volatilities can be implied from derivatives, especially volatility derivatives.

Options imply volatilities. There are different types of derivatives: vanilla options, variance swaps, volatility swaps, etc. Through calibration techniques you'll learn later, you can see how market prices imply volatilities.

5.10 Volatilities affect other variables.

Higher volatility—due to uncertainty in the market—affects liquidity. Similarly, diminishing liquidity can amplify volatility. They affect each other in complex ways. The same can be said for most of the other financial challenges.

5.11 Variances are the diagonal of the covariance matrix.

Within a covariance matrix, the diagonal element contains the individual variances of the securities. So securities that are more volatile will have higher entries. The correlation matrix is scaled, so that can be a more interesting way to see relationships on a more normalized basis.

In short, more volatility means we can expect to be further away from an expected outcome or perhaps even an outcome we never imagined.

6. Statistical Factor 2: Correlation

Let's emphasize some stylized facts about correlation, specifically things that are in the notes for this module and that relate to the previous two modules things. You'll also see these in upcoming modules and even upcoming courses.

Let's consider a dozen properties that are useful to know about correlation from the onset.

- Correlation is bounded and does not have units.
- Correlation does not have a direction.
- Correlation measures linear or monotonic association.
- There are different types of correlation.
- There are different ways to measure correlation.
- Correlation does not imply causality.
- Good regressions have low correlation between predictors.
- Diversified portfolios have low correlation between securities.
- Correlations change over time, especially when stressed.
- Correlation affects over variables (e.g., wrong-way exposure).
- There are alternatives to correlation.

6.1 Correlation is bounded and does not have units.

Correlation is bounded between -1 and 1. Most asset correlations are positive. Extreme correlations (-1 or +1) are rare in practice.

6.2 Correlation does not have a direction.

Correlation does not have a direction. This means that we can swap the order of terms. For example, the correlation of net income and profitability is the same as the correlation of profitability and net income. Note that this is not true for regression. The regression of net income on probability is a different model than the regression of profitability on net income. Regression is directional; correlation is not.

6.3 Correlation measures linear association.

Pearson correlation measures linear association. Recall in our discussion of correlation we also studied Spearman correlation. Spearman measures directional association. Suppose you have two variables, x and y, and they are related through a CAPM-like linear relationship. Graphing y versus x shows a linear relationship. If you switched the relationship to a quadratic, where y contains a squared term in x and a linear term in x, the correlation is not as linear. The higher-order term introduced non-linearity, which would decrease the correlation. You can still have as strong of a relationship, when you include the x-squared term; it is just not linear.

There are examples of nonlinear relationships we have already seen. Recall the graph of a bond's price versus its yield. We get an inverse but convex relationship: the higher the price, the lower the yield. The Pearson correlation of price and yield is not minus one because of convexity. Compare that to the Spearman correlation. Think of Spearman as doing the following: take the correlation of the rank of x and the rank of y. Using ranks instead of values focuses on the direction rather than the size of the move. The Spearman correlation of price and yield is a negative one. In other words, if we increase the price, we lower the yield; we don't do it in a linear way. When the yield is high, small changes in yield cause relatively small changes in price. When the yield is low, small changes in yield cause relatively big changes in price. Those price-yield sensitivities are a result of convexity. When the yield increases, the price increases. Therefore, the ranks of the yield and the ranks of the prices are negatively correlated. Recognize the difference between the Pearson and Spearman correlations: Spearman removes the linear part of correlation and focuses on direction.

6.4 There are different types of correlation.

Depending on the variables we compare, we can calculate different types of correlations. Using two sets of stock returns, we can compute equity correlations. Using two sets of yield, we can compute yield correlations. Using two sets of default probabilities, we can compute default correlations. In credit risk models, we may compute asset correlations.

6.5 There are different ways to measure correlation.

Classical correlation is done using Pearson correlation. There are non-parametric versions like Spearman correlation and Kendall correlation. For time series, we compute autocorrelations and partial correlations. In your studies, be sure to pay attention to any word in front of correlation to understand the specific type of correlation being calculated.

6.6 Correlation does not imply causality.

When we see correlation, we should not assume causality. We don’t want to assume when we see correlated events that one causes the other. There may be confounding variables. Let’s take a classic example. Suppose you find that a child's reading score and shoe size are correlated. Would you say that having bigger feet improves reading ability? The confounding variable here is age. Age and shoe size are correlated; as kids age, their feet grow. Age and reading ability are correlated; as children learn more, they improve their reading. Sometimes, when examining economic variables this is not so obvious because these variables like interest rates, inflation rates, and unemployment rates affect each other in complex ways. There's not a clear cause-and-effect relationship in observational markets. Finance and economics are physical sciences where we are better suited to conduct controlled experiments. Remember, from property one, correlation is not directional, so we can’t get a good sense of what comes first.

6.7 Good regressions have low correlation between predictors.

When you use multiple predictors or factors in regression, you don’t want them to be correlated! If you run a linear regression, you don't want to use two factors that are highly correlated with each other. Suppose you're trying to predict the price or return of a company. Say that you include both the market cap and the market share. Imagine for the selection of stocks in your study, these two variables are highly correlated. The predictors are bumping into each other. You don't want to have too high of a correlation of your predictors. They don't have to have a zero correlation, but you don't want one too high; otherwise, you create problems in the regression. The problem is a violation of one of the regression assumptions about residuals being independent of each other. In some examples you’ll see, we’ll completely eliminate this problem when we use principal components to transform our data into uncorrelated factors. This is a reason to upgrade your math skills so that we're not just measuring correlation.

6.8 Low correlation enhances diversification.

Fifth, just as in our regressors, we prefer to have relatively low correlation in our portfolio. This low correlation facilitates diversification. We'll get to portfolio examples well before the portfolio management course. For now, think that when there is perfect correlation, there is absolutely no diversification. Imagine a vertical line stretching from 1 to -1. Correlation can take any value along that line. As you move down from one to minus one, correlation decreases and diversification increases. If you had a correlation of point nine, you'd still have some diversification, but not much. Recall from Module 3 how diversification benefit is defined. We had an equally weighted portfolio of two stocks, one with a standard deviation of 3 and one with a standard deviation of 8. When the correlation was minus 1, the overall standard deviation was less than the standard deviation of the "safer" one: -2.5. As the correlation increased, the standard deviation increased in a non-linear (i.e., with curvature) way. When the correlation was 1, the standard deviation was just a straightforward average of the two securities. Clearly, as correlation decreases, the diversification benefit increases because the standard deviation decreases. The counterintuitive part is that there is a correlation that makes a combination of the two less risky than either individual security. That's the power of diversification. Incidentally, we call a correlation of -1 a perfect hedge. However, a perfect hedge is not necessarily desirable since it would protect unwanted outcomes but at the cost of wiping out any chance of profits.

Note also you can perform these calculations in your head. The securities have standard deviations of 3 and 8.

With perfect negative correlation, one offsets the other. The difference between eight and three is five: split five in half (with securities equally weighted) and you get a 2.5 standard deviation –remarkably, lower than either security.

With perfect positive correlation, their risks are additive. The sum is 11, making their average 5.5. That is the standard deviation: 5.5.

Due to the relationship between standard deviation and correlation being nonlinear, that's all the calculations you can do in your head. If the correlation is 0—centered between -1 and +1—the standard deviation is NOT 4.0, the average of 2.5 and 5.5. That is because it is nonlinear.

The fact that a combination of securities can be less risky than either security individually, even if both are held long (and not short), is a common interview question: "Can a 2-stock portfolio be less risky than either security in it?" *The answer is "Yes, depending on the correlation!" At minus one, that's possible.*

6.8 Correlations change over time.

Like volatilities, correlations change over time, especially in stressed markets. Ironically, when correlation is the most important—in stressed periods where you would really want diversification and therefore low correlation—correlations may increase. A common example is equity portfolios that contain domestic and foreign stocks. Suppose the average correlation between a domestic stock and foreign stock is 40%. The foreign stocks are there to diversify the portfolio with other economic regions. Suppose the markets are shocked, and the correlation increases because common global macroeconomic factors have affected both equity markets. Now, the average correlation is 70% instead of 40%. Therein lies the problem: the sense of security from the lower correlation and higher diversification is false because in a stressed market the correlation is higher and diversification is lower. As we've seen, lower diversification means the risks are greater. But even if markets aren't stressed, correlations—just like volatilities—can change over time. There's a model that allows you to look at volatility conditionally changing over time. This is a model from the same inventor of GARCH, Rob Engel, and the model for correlations is called the dynamic conditional correlation model. This is just to say that it could be treated as a variable.

6.9 Correlation affects other variables.

Let's revisit credit risk. Recall the formula for expected loss. The formula contains three variables. The expected loss is the product of the probability of default, the loss given default, and the exposure at default. Correlation seems like it doesn't come into the picture, but probability of default and loss given default are positively correlated. Get one wrong and you are likely to get the other one wrong too. In a relatively safe state of the world, there are low default probabilities and low losses given defaults. In a less safe state of the world, there are high default probabilities and relatively high recoveries. Estimate correctly and the expected loss is accurate. Underestimate and the loss is very understated.

6.10 There are alternatives to correlation.

The last property to mention about correlation is that sometimes you don't even want to use it. For example, in a non-stationary environment, you may prefer to find co-integrating variables that can be used instead of correlation. You may want to model joint distributions using copulas; in this case, a correlation value is still needed, but the bivariate relationship is modeled through the lens of a particular copula distribution.

Correlation is not the only way to describe how two variables behave together, but it is often used to discuss how things are related. No one on the financial news discusses cointegration, and even fewer people mention copulas. That's why we're in financial engineering: to enhance our toolkit. These are not difficult topics compared to other mathematics you will see. Once you've had some econometrics and stochastic models and machine learning, your toolkit equips you to eliminate assumptions and model more insightfully.

6.11 Correlation fails.

Correlation is one of the biggest problems in model risk because of the failure to understand joint dependency, particularly in stressed markets. Much of the regulation that came out of the global financial crisis helps to address flawed assumptions about correlation. Among other things, regulation has better prescriptions for large banks, things known as SIFIs, systemically important financial institutions, to perform rigorous stress tests that look at how their liquidity and solvency will stay intact given certain types of correlations that could happen. What would happen if the stock market crashed, liquidity dried up, and GDP tumbled? Stress tests are part of the annual compliance as regulation in many parts of the world now, and banks and other institutions that fail that are out of compliance and are not able to engage in transactions. Volatility can get very high, which can be seen in high VIX values, in data where we reach new highs and new lows, and in headlines about setting records for the most volatile trading day. But correlation is bounded; there is much less of a feel for how a small increase in correlation affects a portfolio.

7. Statistical Factors Combined: Structured Finance Tranches

In our tour of these financial challenges, we categorized the concepts in groups of two. They could occur in any combination. Let us take an example of how correlation and credit risk combine. To do so, let us examine the world of structured finance tranches. For simplicity, suppose you have a collateralized portfolio whose cash flows feed three different tranches:

- Senior tranche
- Junior tranche
- Mezzanine tranche.

The mezzanine tranche absorbs the first 15% loss of the portfolio.

The junior tranche absorbs the next 10% loss of the portfolio.

The senior tranche absorbs the final 5% of the portfolio.

Now, suppose there is LOW default correlation in the collateralized portfolio. Low default correlation means if one security defaults, the others are not affected in any direction. Zero correlation means uncorrelated. If the default probability is low, then we should have a distribution that is skewed. Consider now the situation of a stressed market. Now there is HIGH default correlation in the collateralized portfolio. Suppose it goes to a default correlation of one. Now, all the securities either survive or default.

Let us consider how the default correlation affects these two tranches. The equity tranche is adversely affected by low default correlation. A few defaults and the equity tranche could be wiped out. However, the equity tranche benefits from a high default correlation. Why? Suppose the default correlation is one. This puts the equity tranche in the same risk category as the senior tranche. Either everyone defaults and no one is paid, or no one defaults, in which case equity receives a higher return. The senior tranche benefits from a lower default correlation because it has the subordinate equity tranche top-subsidize the first set of portfolio losses. Thanks to low default correlation, it is highly unlikely that a massive number of securities default, so the senior tranche's payments are protected. However, the senior tranche suffers from a high default correlation. An extremely high default correlation concentrates the risk, putting all tranches into the same position: either all securities survive or they all default. In that case, the senior has the same risk but paid much more for the tranches because it was considered safer. Recall that senior tranches could be rated AAA, but equity tranches are not even rated.

Default correlation changes the risk-reward relationship in the structured finance world. Indeed, we could add volatility to this equation. How? Default correlation plays the role of volatility in the payoffs. This insight is one of the problems in your Group Work Project, where you'll see that the tranches can be thought of as option positions and where the underlying is the portfolio loss. As a hint, think of the type of option or options these positions have.

8. Conclusion

This lesson tackled credit risk, financing, volatility, and correlation. In the next lesson, we'll look at leverage and non-linearity, liquidity and regulation, and model failure and crises.